

Threshold Discovery

Coordination, Crowding, and Minimum Viable Teams

Yohei Nakajima
Untapped Capital

July 2026

Abstract

Some discoveries require a team rather than a single successful action. This changes the role of duplication. With a threshold of one, repeated actions are pure crowding after the first attempt. With a threshold above one, overlap is initially productive because an action does not count until a minimum viable team forms. Too little coordination leaves stranded singletons; too much coordination collapses the portfolio onto one candidate. This paper develops that tension in a finite exact model and synthesizes six bounded components of the Distributed Discovery program. A deterministic planner assigns minimum-size teams to posterior-ranked candidates and obtains the sum of the largest $\min\{\lfloor N/\tau \rfloor, M\}$ posterior masses. Strategic equal-split play need not select that portfolio. Exact small-game enumeration contains weak pure equilibria with zero discovery and distinguishes unilateral equilibrium from pairwise and minimum-team stability. A dynamic extension shows that visible actions can increase herding without increasing discovery in its registered full-credit grid, while a separate threshold-two planner uses both team joining and new-team formation. A mechanism census identifies authoritative, eligibility, contribution, mediation, and matching rules that attain the two-team planner recommendation in bounded common-posterior fixtures, while also preserving equilibrium multiplicity and scope failures. DiscoveryBench v3 and a synthetic-only experiment package connect these results to auditable evaluation and model-derived predictions. Every numerical statement is tied to a checksum-validated immutable run and a claim record. No participants were recruited, no human data were collected, and no experiment was conducted.

Keywords: threshold public goods; minimum viable teams; coordination; crowding; collective search; distributed discovery.

Evidence status. DD-C-0071, DD-C-0072, DD-C-0075, and DD-C-0076 are verified theorem or identity claims. DD-C-0073, DD-C-0074, and DD-C-0077 through DD-C-0087 are bounded exact results with the statuses stated in the claim ledger. DD-C-0088 is a verified Monte Carlo estimate. Synthetic contrasts are predictions, not observations.

1 Introduction

Discovery is often modeled as if one correct attempt were enough. That is the right abstraction for a single searcher opening a file, testing a molecule, or checking a candidate answer. It is incomplete when action requires a team. A field investigation may need two people for safety. An engineering experiment may need several specialties before it can run. A multi-agent system may need a minimum number of independent verifiers before a candidate is accepted. In these settings, spreading every action across a different candidate produces coverage but may produce no executable discovery at all.

result family	claim IDs	evidence category	status
Planner value and payoff identities	DD-C-0071–0072	theorem/identity	verified
Canonical threshold census	DD-C-0073–0074	bounded exact	independently reproduced
Pure and selected mixed equilibrium	DD-C-0075–0076	theorems	verified
Welfare and coalition census	DD-C-0077–0078	bounded exact	independently reproduced
Dynamic attention	DD-C-0079–0082	proof/exact/negative	verified or reproduced
Minimum-team mechanisms	DD-C-0083–0086	bounded exact	verified
DiscoveryBench v3	DD-C-0087	bounded exact	independently reproduced
Synthetic predictions	DD-C-0088	Monte Carlo estimate	verified
Human behavior	none	empirical	not studied

Table 1: Evidence status attaches to statements, not to the paper as a whole.

Source runs: 20260722T021526Z_DD-016_00271ff8_123b2809e3, 20260722T024032Z_DD-017_033452f6_3d2c74fdfb, 20260722T043713Z_DD-015_92d53ac1_0e7cf1ec0a, 20260722T044453Z_DD-015_34bc4379_33e1da478b, 20260722T051847Z_DD-018_a193f602_3b3ddac173, 20260722T054447Z_DD-010_d265e480_6930915b02, 20260722T061958Z_DD-011_5743ccba_19b6517655; claims: DD-C-0071–DD-C-0088; generator: distributed_discovery.papers.build_threshold_discovery; input SHA-256 prefixes: 473fb4ffe40f, d2dac2854a3f, d658f36d857e, 02d4c2d76cbc, 0a31654fd47a, 0439b211955c, 6d28c36cc72e, d1bd94a7dd79, f1eb762c73c9, 49a36beb1f03, d2c936e7ed55.

The organizational problem becomes a portfolio problem with lumpy capacity. Let N agents choose among M candidates, and let candidate j count only if at least τ agents choose it. A planner cannot simply maximize the number of distinct actions. The planner must build viable teams. Yet forming teams does not make concentration universally good. After the threshold is met, more agents on the same candidate add no discovery value in the frozen model. The right unit of diversification is therefore not the individual action but the minimum viable team.

That observation creates three separate questions. First, what portfolio would a planner choose when the posterior over candidates is known? Second, what allocations can arise when agents choose strategically and share a discovery prize? Third, which institutions can support a high-value team portfolio under clearly stated information, authority, transfer, and deviation assumptions? Treating those questions as one would conceal the difference between an optimum, an equilibrium, and an implementation result.

The planner problem has a simple exact solution. No allocation can open more than $\lfloor N/\tau \rfloor$ candidates, and no more than M candidates exist. Opening the posterior-ranked candidates is optimal. An attaining allocation places exactly τ agents on each selected candidate before assigning any leftovers. The value is

$$V^P(\pi, N, \tau) = \sum_{j=1}^{\min\{\lfloor N/\tau \rfloor, M\}} \pi_{(j)}, \quad (1)$$

where $\pi_{(1)} \geq \dots \geq \pi_{(M)}$. This is an existence and value theorem, not uniqueness: posterior ties and leftover agents may generate many optimal assignments (DD-C-0071).

The strategic problem is less forgiving. When a target candidate opens, the unit prize is split among agents on that candidate. An agent then compares its share at the current occupancy with the share after a unilateral move. A pure occupancy is a weak Nash equilibrium exactly when none of those finite moves improves payoff (DD-C-0075). Weakness matters. If no candidate has reached the threshold, one agent may be unable to create value alone. The resulting zero payoff indifference can stabilize an allocation with zero discovery even while the planner value is positive.

The paper also separates one symmetric mixed construction from the full equilibrium correspondence. Uniform independent mixing over tied posterior modes is always a symmetric mixed equilibrium for $\tau \geq 2$ in the frozen game, because a supported mode offers the same positive payoff and an unsupported unilateral move cannot form a team. At $\tau = 1$, an unsupported candidate can instead pay its posterior mass immediately, so a precise outside payoff condition is required (DD-C-0076).

The selected mixture is useful for comparison but is not a universal theory of equilibrium selection.

The synthesis adds time and institutions. In the dynamic model, agents observe a shared clue, a private clue, and possibly earlier actions. Under full duplicate credit, choosing a posterior mode at every reached information set is a sequential Bayesian equilibrium (DD-C-0079). Exact bounded evaluation shows that visible actions need not improve discovery and can increase herding (DD-C-0081). Under a separately registered threshold-two planner, good policies sometimes start a new singleton, sometimes join one, and sometimes reinforce an already viable team (DD-C-0082). Timing does not erase the coordination- crowding tradeoff; it makes the relevant action depend on the current portfolio.

Finally, institutions can change who is able or rewarded to form which team. A ten-mechanism census distinguishes authority, recommendation, eligibility, contribution rewards, and matching. Eight mechanisms attain the planner portfolio in each of five common-posterior fixtures, while universal pooling and sole-team rescue collapse to one candidate and do not (DD-C-0083). Some strategic rules support the recommendation against unilateral and specified two-person deviations, but retain many other pure equilibria (DD-C-0086). Recommendation support is therefore not equilibrium selection.

The contribution is a disciplined map of the threshold-discovery problem. It connects an exact planner theorem, bounded equilibrium and stability censuses, dynamic attention, mechanism design, benchmark integration, and synthetic predictions without upgrading any component beyond its evidence. The paper makes no verified novelty claim. It is a working paper, not a DOI-bearing, submitted, or peer-reviewed publication.

2 Related literature

Threshold discovery sits near several established literatures while using a different objective. Threshold public-good and participation games study when enough contributors coordinate to provide a good; early analyses emphasize the sharp strategic effects of minimum participation requirements [Palfrey and Rosenthal, 1984, Diekmann, 1985]. Here the public outcome is not a single project. Agents can form several candidate-specific teams, and the planner values the posterior mass covered by the resulting portfolio.

Congestion games study how payoffs change with resource occupancy [Rosenthal, 1973]. Equal splitting in the present model creates a form of crowding after a candidate is viable, but occupancy below the threshold is complementary rather than congested. The same additional teammate can be essential when occupancy rises from $\tau - 1$ to τ and redundant when it rises from τ to $\tau + 1$. That sign change is the organizing feature of the paper.

Coalition formation and hedonic grouping ask which collections of agents are stable under different deviation concepts [Bogomolnaia and Jackson, 2002]. Cooperative-game concepts such as strong equilibrium and coalition-proofness also distinguish individual from group deviations [Aumann, 1959, Bernheim et al., 1987]. The bounded census below uses narrower registered objects: weak pure Nash, strict-member pair deviations, and exact-size- τ strict-member deviations. It does not relabel those checks as strong or coalition-proof equilibrium.

Information aggregation and organizational design study how shared evidence changes action and coordination [Marschak and Radner, 1972, Hellwig and Veldkamp, 2009, Myatt and Wallace, 2012]. The Distributed Discovery framework emphasizes that better group ranking can coexist with a worse portfolio of actions [Nakajima, 2026, Wang et al., 2026]. Thresholds add a complementary constraint: maximal action diversity can also be useless if no candidate receives enough effort to open.

Collective search and scientific division of labor ask how groups balance exploitation and ex-

ploration [March, 1991, Kitcher, 1990, Zollman, 2007]. The finite model contributes a transparent capacity rule. When each discovery requires τ agents, the planner has at most $\lfloor N/\tau \rfloor$ viable search units. This supplies a compact benchmark for more elaborate dynamic, heterogeneous, or costly-search models.

Mechanism design adds another distinction. A central assignment can implement an allocation by authority. A mediator can recommend actions. A token or right can condition prize eligibility. A contribution rule can reward membership in an exactly minimal team. These are different institutional objects even when their recommended allocations coincide. The paper therefore reports observed information, action observability, commitment, transfers, subsidy, identity selection, and deviation class rather than treating “coordination” as a single intervention.

Boundary. The literature discussion locates the model; it does not certify novelty. The formal and numerical claims of this paper are only those linked to the repository claim ledger and immutable runs.

3 Threshold discovery model

There is a finite candidate set $\mathcal{M} = \{1, \dots, M\}$ and one hidden target θ . Conditional on the available evidence, the posterior is $\pi = (\pi_1, \dots, \pi_M)$ with positive entries summing to one. There are N agents. Each agent chooses one candidate. Let n_j be the occupancy of candidate j , so $\sum_j n_j = N$.

A candidate is viable when $n_j \geq \tau$. Discovery is the event that the true target is among the viable candidates. Conditional expected discovery is therefore

$$D(n; \pi, \tau) = \sum_{j=1}^M \pi_j \mathbf{1}\{n_j \geq \tau\}. \quad (2)$$

There is no additional value from occupancy above τ , no execution failure once the threshold is met, and no cost in the planner objective. Those assumptions make the capacity logic exact. They also define the boundary of the result.

For strategic analysis, the target candidate carries a unit prize divided equally among agents who selected it, provided it is viable. An agent on candidate j obtains posterior expected payoff

$$u_j(n) = \begin{cases} \pi_j/n_j, & n_j \geq \tau, \\ 0, & n_j < \tau. \end{cases} \quad (3)$$

A unilateral move from j to k compares $u_j(n)$ with $\pi_k/(n_k + 1)$ if $n_k + 1 \geq \tau$ and with zero otherwise. This finite comparison is the basis of DD-C-0075.

The mixed-action identity makes the same threshold visible probabilistically. Suppose each of the other $N - 1$ agents independently chooses j with probability x_j , and let $K_j \sim \text{Binomial}(N - 1, x_j)$. Choosing j has expected payoff

$$U_j(x) = V \pi_j \mathbb{E} \left[\frac{\mathbf{1}\{K_j \geq \tau - 1\}}{K_j + 1} \right]. \quad (4)$$

At $\tau = 1$, the share factor equals $[1 - (1 - x_j)^N]/(Nx_j)$, with the exact zero limit equal to one. At $\tau = 2$, the $K_j = 0$ term is removed and the zero limit is zero. These are verified identities, not an equilibrium characterization (DD-C-0072).

Three benchmarks recur. Private clue-following sends each agent to the label named by its clue. Common deterministic mode pools the clues and sends everyone to a selected posterior mode. Registered tied-mode mixing makes independent uniform draws over all posterior modes. The planner allocates minimum-size teams to the highest posterior candidates. The first three rules turn evidence into actions; the fourth is an allocation frontier. None should be called the other.

4 Minimum viable teams

4.1 The planner theorem

Let $K = \min\{\lfloor N/\tau \rfloor, M\}$. Any allocation opening q candidates uses at least $q\tau$ agents, so $q \leq K$. Among all sets with at most K candidates, the largest posterior mass is the sum of the K largest entries. Assigning τ agents to each of those candidates attains the bound, and any leftover agents can be placed without closing an opened candidate. This proves Equation 1 (DD-C-0071).

The proof yields an institutional rule of thumb: build the smallest teams that can act, rank opportunities by posterior value, and diversify at the team level. It does not say that every optimum has exactly the same occupancy. Leftovers may move freely when they do not change viability. Posterior ties may change which candidate labels are selected. If value continues to rise with team size, execution can fail, or team members have heterogeneous skills, the minimum-team construction need not remain optimal.

4.2 The canonical phase diagram

DD-016 freezes a canonical signal model with $M = 16$, $N = 8$, conditionally independent clues, and clue accuracy $1/5$. Exact fraction arithmetic exhausts 490,314 labeled clue-count vectors. A materially different occupancy-orbit enumeration reconstructs the same aggregates. Figure 1 and Table 2 are generated directly from that immutable output (DD-C-0073 and DD-C-0074).

At $\tau = 1$, individual diversity is useful because every distinct correct action can discover. At $\tau = 2$, many private or independently mixed actions remain stranded as singletons. Common mode is still too concentrated, but the gap between uncoordinated and coordinated portfolios changes. From $\tau = 3$ through 8, common mode and the registered tied-mode mixture exceed private clue-following in the frozen fixture. Coordination becomes necessary to make any candidate viable often enough.

The planner remains the appropriate frontier because it chooses whole teams. For $\tau = 1$ through 4, it strictly diversifies beyond common mode and earns a higher discovery value. For $\tau = 5$ through 8, only one viable coalition can form with eight agents, so planner diversification cannot open a second candidate and the planner ties common mode. The apparent convergence is a hard capacity constraint, not evidence that universal concentration is generally optimal.

The phase diagram also clarifies why “crowding” cannot be measured by raw collisions alone. Some overlap is necessary: τ agents must coincide before a candidate counts. Excess overlap is occupancy beyond what is needed to form the viable teams in the selected portfolio. Failed subthreshold attempts and excess members of already viable teams are different losses and demand different interventions.

4.3 Posterior ranking and action diversity

Posterior ranking determines which teams the planner would form, not how private agents come to share the same ranking. A common posterior can be an input to an assignment problem. Dispersed

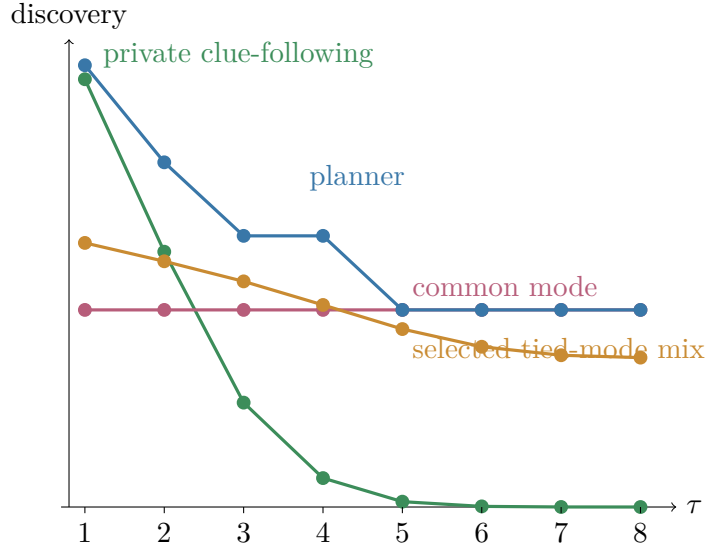


Figure 1: Exact canonical threshold phase diagram for $M = 16$, $N = 8$, and clue accuracy $1/5$. Common mode and the registered tied-mode mixture trail private clue-following at thresholds one and two, then exceed it at thresholds three through eight. The deterministic planner remains the comparison frontier.

Source runs: 20260722T021526Z_DD-016_00271ff8_123b2809e3; claims: DD-C-0073, DD-C-0074; generator: distributed_discovery.papers.build_threshold_discovery; input SHA-256 prefixes: 473fb4ffe40f.

private signals create an additional information-design problem. The exact canonical census holds the signal structure fixed and compares named action rules; it does not prove that pooling information is always desirable or always harmful.

This separation matters operationally. An organization may have a high-quality central ranking but lack assignment authority. It may have assignment authority but unreliable execution. Or it may have decentralized signals that cannot be truthfully aggregated. Equation 1 answers the allocation problem conditional on the posterior. It does not solve the upstream acquisition and reporting problems.

5 Strategic equilibrium and selection

5.1 Pure occupancy equilibrium

The pure equilibrium characterization is deliberately direct. An occupancy is a weak Nash equilibrium if and only if every occupied agent weakly prefers the current candidate payoff in Equation 3 to every unilateral destination payoff after adding one agent there. There are no other pure unilateral deviations in the frozen action space, so the condition is necessary and sufficient (DD-C-0075).

Weak equilibrium preserves economically meaningful coordination failures. Suppose $\tau = 2$ and every agent is a singleton. Each receives zero. A unilateral move may create one viable pair, but whether it is profitable depends on the destination posterior and resulting split. In other states, no single move can create a team and all zero-payoff actions are tied. Removing such equilibria by assuming strict tie-breaking would change the game rather than merely simplify the analysis.

DD-017 registers eight positive rational posterior fixtures with M between two and four, N between

τ	private	common	mixed	planner	viable teams	planner relation
1	0.8322	0.3835	0.5138	0.8594	1.7171	strictly higher
2	0.4967	0.3835	0.4780	0.6706	1.4198	strictly higher
3	0.2031	0.3835	0.4390	0.5275	1.1876	strictly higher
4	0.0563	0.3835	0.3931	0.5275	0.9875	strictly higher
5	0.0104	0.3835	0.3462	0.3835	0.8049	ties common
6	0.0012	0.3835	0.3118	0.3835	0.6748	ties common
7	0.0001	0.3835	0.2953	0.3835	0.6128	ties common
8	0.0000	0.3835	0.2906	0.3835	0.5956	ties common

Table 2: Displayed decimals summarize exact rational values. Exact fractions in the immutable source determine every comparison.

Source runs: 20260722T021526Z_DD-016_00271ff8_123b2809e3; claims: DD-C-0073, DD-C-0074; generator: distributed_discovery.papers.build_threshold_discovery; input SHA-256 prefixes: 473fb4ffe40f.

quantity	count	interpretation
Registered games	160	bounded exact census
Occupancy states	3728	primary enumeration
Labeled profiles	87216	independent reproduction
Games with zero worst-equilibrium discovery	52	DD-C-0077
Games with no pairwise-strict-stable equilibrium	8	DD-C-0078
Games with no exact-size- τ -stable equilibrium	35	DD-C-0078
Registered tied-mode failures	21	all at $\tau = 1$

Figure 2: The equilibrium problem is not a single selection rule. Weak pure Nash, one symmetric mixture, pairwise blocking, and exact-size minimum-team blocking are separate registered objects.

Source runs: 20260722T024032Z_DD-017_033452f6_3d2c74fdff; claims: DD-C-0075, DD-C-0076, DD-C-0077, DD-C-0078; generator: distributed_discovery.papers.build_threshold_discovery; input SHA-256 prefixes: d658f36d857e.

two and six, and every integer threshold from one to N . It exhausts 3,728 occupancy states. An independent labeled-profile enumerator visits 87,216 profiles, deduplicates their occupancies, and reconstructs the same equilibrium correspondence.

In 52 of the 160 games, the worst weak pure equilibrium has zero discovery even though the planner value is positive. Under the registered positive-over-zero convention, the pure price of anarchy is infinite. This is a bounded exact result, not a theorem that every threshold game has an inefficient equilibrium (DD-C-0077).

5.2 One symmetric mixture

If m candidates tie for the highest posterior and every agent independently mixes uniformly over those modes, support payoffs are equal. For $\tau \geq 2$, an unsupported unilateral deviation cannot form a viable team and pays zero, while a supported action has positive probability of joining enough other agents. The mixture is therefore a symmetric mixed Nash equilibrium.

At $\tau = 1$, an unsupported action immediately opens its candidate. The support payoff must weakly exceed every unsupported posterior mass:

$$\pi_{\max} \frac{1 - (1 - 1/m)^N}{N/m} \geq \max_{k \notin S} \pi_k. \quad (5)$$

The condition can fail. All 21 failures in the registered census occur at $\tau = 1$. This theorem verifies one symmetric construction but neither characterizes all mixed equilibria nor supplies a learning or selection process (DD-C-0076).

That qualification is essential when interpreting the canonical DD-016 tied-mode curve. The curve is a registered comparison rule. For thresholds two and above it is also one equilibrium under the frozen equal-split game, but it does not replace the full pure or mixed equilibrium correspondence. An institutional recommendation must say which selection process, if any, makes that curve behaviorally relevant.

5.3 Coalitional stability

Unilateral equilibrium does not settle whether a team can form by moving together. DD-017 therefore asks two bounded questions. A pair blocks only if a joint action change makes both named members strictly better. An exact-size- τ coalition blocks only if a joint change makes every named member strictly better. The two concepts coincide in coalition size when $\tau = 2$ but retain separate definitions.

Eight registered games have no pairwise-strict-stable weak pure equilibrium, and 35 have no exact-size- τ -strict-stable weak pure equilibrium (DD-C-0078). These counts are not nested comparative statics. Exact-size stability is not monotone in coalition size, and neither result tests arbitrary coalitions, transfers, weak Pareto improvements, strong equilibrium, or coalition-proofness.

The substantive lesson is methodological. A mechanism can pass unilateral obedience while failing a team deviation. It can pass a minimum-team deviation while retaining other bad equilibria. And it can be authoritative, making deviation tests not applicable rather than passed. Those labels must remain visible in any mechanism comparison.

6 Dynamic attention

Static occupancy hides the order in which agents observe and act. DD-015 freezes a three-target sequential model with two or three agents, one shared clue, private clues, visible or hidden action history, and either fixed-budget or stopping feedback. The reward in the autonomous game is full duplicate credit: an agent receives one whenever its own action is correct, regardless of what others do.

Under that reward, an agent’s expected payoff from action a is exactly its posterior probability that $\theta = a$. Later actions do not change that payoff. Choosing a posterior mode at every reached information set is therefore a pure sequential Bayesian equilibrium, with the registered lowest-label tie-break (DD-C-0079). This proposition does not extend to the equal-split threshold game or to contribution and rescue rewards.

The exact grid contains 64 objective rows. The common-payoff planner strictly exceeds autonomous full-credit play in 38. Autonomous agents follow a previous action with positive probability in 32 rows and lean against a repeated prior action with positive probability in 50. Stopping reduces expected actions relative to fixed budget in every one of 32 parameter cells (DD-C-0080).

The visibility result is a preserved bounded null. Across 32 fixed-budget cells, visible-action inference improves neither discovery nor expected action dispersion relative to a history-hidden Bayesian baseline. It strictly lowers both in 18 cells and ties in 14 (DD-C-0081). Earlier actions carry information, but under full duplicate credit each agent optimizes individual correctness. Informational value and portfolio value can therefore move in opposite directions.

Stopping feedback is different because continuation reveals failure as well as the earlier action. The fixed-budget visibility comparison excludes this extra signal. Combining the two would make the negative result appear broader than the registered design.

dynamic object	count	evidence boundary
Baseline parameter cells	32	exact
Baseline labeled paths	5184	independently reproduced
Planner-better objective rows	38	bounded result
Stopping reduces expected actions	32	all baseline cells
Visibility increases fixed-budget herding	18	bounded negative result
Visibility increases fixed-budget discovery	0	none
Threshold-two parameter cells	16	separate planner-only extension
Threshold-two labeled paths	3888	independently reproduced

Figure 3: Dynamic attention results preserve two distinct games. The autonomous baseline has full duplicate credit; the threshold-two extension is planner-only and does not establish a threshold-game equilibrium.

Source runs: 20260722T043713Z_DD-015_92d53ac1_0e7cf1ec0a, 20260722T044453Z_DD-015_34bc4379_33e1da478b; claims: DD-C-0079, DD-C-0080, DD-C-0081, DD-C-0082; generator: distributed_discovery.papers.build_threshold_discovery; input SHA-256 prefixes: 02d4c2d76cbc, 0a31654fd47a.

The threshold-two extension is separately registered and planner-only. With three sequential agents and $\tau = 2$, stopping preserves the fixed-budget optimal discovery value while strictly reducing expected actions in all 16 parameter cells. Across its 32 objective rows, optimal policies use each of five action categories with positive probability: start a singleton, join a singleton, join a viable team, follow the shared clue, and oppose it (DD-C-0082).

Joining a viable team after discovery is already secured can occur under fixed budget because remaining actions must still be assigned. Its presence is not evidence of decentralized over-crowding. Conversely, starting a singleton is not automatically wasteful: it can preserve an option for a later agent. The dynamic state determines whether another action completes a team, preserves a portfolio, or becomes redundant.

7 Mechanisms for team formation

DD-018 fixes three candidates, four agents, threshold two, and five weakly descending common posterior fixtures. The planner wants two teams of two on the top candidates. Ten institutions are registered with explicit information, observability, commitment, transfers, subsidy, anonymity, identity selection, and deviation class.

Central assignment and random matching are authoritative after assignment. Their planner performance comes from control, so post-assignment unilateral obedience is not applicable. Team tokens and exclusive coalition rights make prize eligibility depend on identity or charter. A correlated mediator sends private recommendations without binding action. Universal pooling uses ordinary threshold-adjusted equal splitting at a common lexicographic mode. Sole-team rescue pays only when exactly one viable team exists. Marginal coalition contribution pays an exactly minimal viable team. A threshold-adjusted equal-split recommendation and a pairwise matching market complete the registry.

Across 50 mechanism-fixture rows, 40 recommendations attain the two-team planner value. Central assignment, random matching, team tokens, exclusive coalition rights, correlated mediation, marginal coalition contribution, threshold-adjusted equal split, and the pairwise matching market attain it in all five fixtures. Universal pooling and sole-team rescue attain it in none (DD-C-0083).

Among 35 rows where unilateral action changes are meaningful, 31 recommendations are strictly

mechanism	planner rows	strict unilateral	pair stable	pure multiplicity range
central assignment	5/5	0/5	0/5	n/a
random matching	5/5	0/5	0/5	n/a
team tokens	5/5	5/5	5/5	25–25
exclusive coalition rights	5/5	5/5	5/5	25–25
correlated mediator	5/5	3/5	4/5	3–21
universal pooling	0/5	5/5	1/5	3–21
sole team rescue	0/5	5/5	5/5	3–3
marginal coalition contribution	5/5	5/5	5/5	21–21
threshold adjusted equal split	5/5	3/5	4/5	3–21
pairwise matching market	5/5	0/5	4/5	1–6

Table 3: Exact five-fixture mechanism census. Authoritative and binding-within-pair rows use not-applicable labels where unilateral obedience is not a meaningful test. Common-posterior input also makes report truthfulness not applicable.

Source runs: 20260722T051847Z_DD-018_a193f602_3b3ddac173; claims: DD-C-0083, DD-C-0084, DD-C-0085, DD-C-0086; generator: distributed_discovery.papers.build_threshold_discovery; input SHA-256 prefixes: 6d28c36cc72e.

obedient. Twenty-nine withstand every registered strict-member pair deviation and every exact-size-two τ -player deviation. All 50 rows satisfy nonnegative participation, weak prize-budget balance, and zero external subsidy (DD-C-0084). Report truthfulness is not applicable because the common posterior is an input, not elicited from private reports.

The negative mechanism result is as informative as the successes. Universal pooling and sole-team rescue place all four agents on the top candidate and earn strictly less discovery than the two-team planner in every fixture. Universal-pooling collapse is strictly unilaterally obedient in all five fixtures but pairwise stable in only one. Sole-team-rescue collapse is strictly unilaterally obedient and pairwise stable in all five (DD-C-0085). Even a locally stable reward can preserve global concentration.

Team tokens, exclusive coalition rights, and marginal coalition contribution support the two-team recommendation with strict unilateral obedience, no registered strict-member pair or size-two block, weak budget balance, and no subsidy in every fixture. Yet their labeled pure-Nash multiplicities are 25, 25, and 21 in every fixture (DD-C-0086). Supporting one good recommendation does not imply uniqueness or a process that selects it.

The mechanism census suggests an architecture rather than a universal winner. Where authority is legitimate, assignment can directly build minimum teams. Where action remains voluntary, eligibility or marginal contribution can alter the private payoff from joining the intended team. Where identities should be anonymous ex ante, a mediator or random matching can distribute roles. Each option trades autonomy, information requirements, and selection risk.

Boundary. The mechanism evidence is a five-fixture exact common-posterior census. It does not establish private report truthfulness, arbitrary coalition stability, coalition-proofness, endogenous source acquisition, or a general implementation theorem.

8 Benchmark and synthetic predictions

Program V4 integrates the exact results into DiscoveryBench v3 and a versioned synthetic experiment package. These artifacts serve different purposes. The benchmark checks whether capability-declared protocols reproduce registered exact task vectors. The experiment package maps model contrasts into treatments, outcomes, hypotheses, and conditional power calculations. Neither is a field evaluation.

DiscoveryBench v3 preserves every v1 and v2 exact vector while extending the opt-in registry to 24 tasks, 29 protocols, and 39 metrics. Thirty-six compatible pairs reproduce their exact registered

integration object	count	status
DiscoveryBench v3 tasks	24	exact registry
DiscoveryBench v3 protocols	29	capability declared
DiscoveryBench v3 metrics	39	no composite score
Compatible exact pairs	36	independently reproduced
Explicit exclusions	660	not imputed
Synthetic treatment cells	37	no human data
Synthetic power rows	1680	seeded Monte Carlo
Retained large-sample failures	644	published negative calibration

Figure 4: Benchmark and experimental integration preserve capability exclusions and calibration failures. The synthetic package supplies conditional predictions only; it is not behavioral or empirical evidence.

Source runs: 20260722T054447Z_DD-010_d265e480_6930915b02, 20260722T061958Z_DD-011_5743ccba_19b6517655; claims: DD-C-0087, DD-C-0088; generator: distributed_discovery.papers.build_threshold_discovery; input SHA-256 prefixes: d1bd94a7dd79, f1eb762c73c9.

hypothesis	registered outcome	N	estimated power
H15	discovery	960	1.000000
H16	threshold portfolio efficiency	960	0.999000
H17	viable team count	960	0.918000
H18	discovery	960	0.948000
H19	action savings	960	1.000000
H20	discovery	960	0.999000

Table 4: Program V4 synthetic predictions in the favorable rational-response scenario. Power is conditional on declared effects and a normal approximation; it is not evidence that any effect occurs in people.

Source runs: 20260722T061958Z_DD-011_5743ccba_19b6517655; claims: DD-C-0088; generator: distributed_discovery.papers.build_threshold_discovery; input SHA-256 prefixes: d2c936e7ed55, 49a36beb1f03.

values. The other 660 of 696 candidate pairs remain explicit exclusions. Missing metrics are not imputed, hidden target-state access is rejected, and no universal composite score is defined (DD-C-0087).

Synthetic Experiment v3 preserves the earlier registries and shared-seed v2 power rows while extending the package to 37 treatment cells, 20 hypotheses, 23 outcomes, 14 response scenarios, 1,184 balanced synthetic assignments, and 1,680 power rows with 1,000 replications each. Among 840 registered rows at sample sizes 640 or greater, 644 remain below the 0.80 calibration threshold. Every failure is retained (DD-C-0088).

The threshold and dynamic treatments are predictions about contrasts, not claims about people. The favorable rational-response scenario at sample size 960 gives estimated powers 1.000, 0.999, 0.918, 0.948, 1.000, and 0.999 for H15 through H20. Those values depend on the declared assumed effects, normal sampling approximation, Bonferroni family alpha, response multiplier, and fixed random streams. A different behavioral response model can produce much lower power.

The experimental design is intentionally honest about failure. A power row below 0.80 is not evidence that a treatment has no effect, just as a high synthetic power estimate is not evidence that it does. The table is a planning calculation conditional on synthetic assumptions. No participants were recruited. No human data were collected. No experiment was conducted.

The benchmark likewise does not rank arbitrary agents. It evaluates only compatible task-protocol-metric combinations. A protocol that lacks an input capability is excluded rather than assigned a poor score; a protocol with forbidden target-state access is excluded rather than rewarded. This keeps the exact matrix interpretable and prevents a convenient aggregate from hiding incommensurable tasks.

9 Institutional implications

9.1 Form teams, then diversify teams

The planner theorem provides the simplest design principle. When success needs τ agents and additional members add no value, the scarce resource is the number of viable teams, not the number of individuals. Organizations should first make teams viable and then place successive teams on posterior-ranked candidates. This is the threshold counterpart of action diversification.

The principle is conditional. It applies when the posterior is available, team members are interchangeable, execution is deterministic at the threshold, and there is no scale value beyond it. A surgical team, research group, or model ensemble may violate all of those assumptions. In such cases, the minimum-team portfolio remains a baseline against which the value of heterogeneity, redundancy, and scale can be measured.

9.2 Measure stranded effort and excess crowding separately

An occupancy below τ is a stranded attempt in the frozen one-shot model. Occupancy above τ is excess crowding if those agents could instead complete another valuable team. Aggregate concentration measures can mix the two. A system can have low concentration yet no viable team, or high concentration with exactly the overlap needed for execution.

Useful operational metrics therefore include the number of viable candidates, failed subthreshold attempts, excess members on viable candidates, expected largest crowd, and posterior mass opened. DD-016 reports all of these as exact objects. DiscoveryBench v3 keeps the capabilities and metric definitions separate rather than compressing them into one score.

9.3 Do not confuse support with selection

A recommendation can be obedient and still coexist with many other equilibria. The mechanism census makes multiplicity visible. A deployment claim should name not only a supported profile but also the assumed selection process: authority, defaults, repeated adjustment, mediation, matching, or some other rule. Without that process, “implements” may mean only that one good equilibrium exists.

The same warning applies to the registered tied-mode mixture. It is one symmetric equilibrium for thresholds above one, but it is not the complete equilibrium set. Comparing its discovery value with the planner is informative about that selection rule and nothing broader.

9.4 Use dynamic observability deliberately

Visible actions can convey information and simultaneously induce correlated behavior. The DD-015 null shows no discovery or dispersion gain from visibility in the registered fixed-budget full-credit grid.

This does not justify hiding all actions. It just rejects the assumption that more observed history must improve portfolio discovery.

Stopping rules add a different signal: continuing reveals that earlier actions failed. In the exact dynamic studies, stopping reduces expected actions while preserving the relevant planner value in the threshold-two extension. Systems should distinguish action visibility, outcome visibility, and continuation feedback rather than bundle them as “transparency.”

9.5 Match rewards to contribution and rights

Equal splitting can reward joining an already popular candidate. Eligibility rights and marginal contribution can instead attach reward to the intended team structure. The bounded mechanism results show that these devices can support the two-team planner profile without external subsidy in the registered fixtures. They also introduce identity, observability, and enforcement demands.

No rule is free. Tokens require allocation and verification. Coalition rights name privileged identities. Contribution rewards require occupancy observability and a definition of the minimal team. Mediation requires trusted recommendations. Assignment requires authority. The institutional choice should report these inputs and not only the achieved discovery number.

10 Limitations

The model has one hidden atomic target and a normalized posterior. Actions name candidates directly. Agents are homogeneous in capacity, each takes one action, and a candidate opens deterministically at a common integer threshold. There is no partial progress, execution failure, false positive cost, learning spillover, or value from effort above the threshold. Equation 1 is exact inside that model and should not be exported without restating it.

The strategic game uses a unit target prize split among agents at a viable true candidate. Other reward rules produce other equilibria. The pure equilibrium theorem covers unilateral deviations; the bounded stability census covers only strict-member pairs and exact-size- τ coalitions. Neither is a strong- equilibrium or coalition-proof analysis.

The canonical threshold census uses one $M = 16$, $N = 8$, $p = 1/5$ signal fixture. Its phase changes are exact for that fixture but not universal comparative statics. The equilibrium census uses 160 small rational games. Counts such as 52, 8, and 35 are bounded observations, not asymptotic laws.

DD-015 uses three targets and two or three sequential agents. Its autonomous equilibrium relies on full duplicate credit and reached information sets. Its visibility null excludes stopping feedback from the comparison. Its threshold- two extension is planner-only and must not be described as decentralized equilibrium evidence.

DD-018 takes a common posterior as input, so it cannot test report truthfulness. Its threshold is two and its coalitional checks are correspondingly small. The mechanism names denote exact registered rules, not every institution that could use the same ordinary-language label. The finding that sole-team rescue fails in five fixtures is not a theorem that rescue payments generally fail.

DiscoveryBench v3 is an exact internal benchmark, not a hosted leaderboard or universal measure of agent quality. Synthetic Experiment v3 uses declared effects and response scenarios. Its Monte Carlo estimates are conditional planning evidence only. The 644 retained failures are part of the design record, not behavioral null results.

Most importantly, there is no empirical validation. No participant, organization, laboratory, or

deployed multi-agent system was observed. Human deployment would require separate ethics, consent, privacy, retention, identification, pilot, and governance review. The paper supplies hypotheses and auditable finite benchmarks, not causal evidence.

11 Conclusion

Threshold discovery replaces a simple diversity maxim with a portfolio rule. When a candidate requires a minimum viable team, some duplication is necessary. The planner forms the smallest viable teams on posterior-ranked candidates. Uncoordinated diversity can strand effort, while common-mode concentration can waste team capacity. The relevant design objective is disciplined overlap.

Strategic behavior complicates that planner rule. Weak equilibria can have zero discovery. One natural symmetric mixture is only one selection. Pair and minimum-team stability answer different questions. Dynamic observability can increase herding without increasing discovery. Mechanisms can support a good portfolio while retaining many other equilibria.

The practical message is correspondingly precise: identify the minimum viable team, measure both stranded attempts and excess crowding, diversify whole teams, and state the authority and reward system that makes the portfolio credible. Benchmark capability and calibration failures should remain visible. None of these prescriptions is an empirical finding until tested under a separately authorized design.

Threshold Discovery is therefore a bridge between information and institution. Evidence ranks opportunities; team technology determines the granularity of action; incentives and authority determine which portfolio is actually formed; feedback changes what later agents know. A discovery architecture is complete only when all four are specified.

A Evidence and reproduction map

The paper uses seven immutable run IDs. DD-016 supplies the planner proof audit, payoff identities, canonical metrics, and eight-row phase diagram. DD-017 supplies the 160-game equilibrium and coalition census. DD-015 supplies one baseline dynamic run and a separately labeled threshold-two extension. DD-018 supplies the ten-mechanism, five-fixtured census. DD-010 and DD-011 supply the versioned benchmark and synthetic experiment integration.

Each generated asset prints its contributing claim IDs, run IDs, generator, and input checksum prefixes. Full checksums are stored in ‘generated/provenance.json’. The builder first reads each immutable manifest, requires passed validation and exit status zero, resolves the named output, and compares its SHA-256 with the manifest. A missing or changed source stops the build.

The claim audit extracts every token matching DD-C- followed by four digits from the manuscript and generated assets, then resolves it against `claims/claims.yml`. Citation keys are extracted from the manuscript and resolved against the generated bibliography. Required sections and every generated input are checked before compilation.

Tectonic compiles the manuscript twice with `SOURCE_DATE_EPOCH` fixed to the DD-016 run start. The builder rejects nonzero exit status, undefined references, undefined citations, and overfull boxes. The two PDF byte streams must have the same SHA-256. Poppler supplies the page count and renders every page for visual review. The committed validation record binds the source inputs, generated assets, page count, and final PDF checksum.

B Exact claims by component

DD-C-0071 is the general finite planner theorem. DD-C-0072 is the finite binomial payoff identity. DD-C-0073 and DD-C-0074 are independently reproduced canonical computational results. Their exact fractions, not displayed decimals, determine all labels.

DD-C-0075 characterizes weak pure Nash occupancy in the frozen threshold game. DD-C-0076 verifies one symmetric tied-mode mixture and its threshold-one boundary. DD-C-0077 and DD-C-0078 report the bounded welfare and coalition censuses. The latter do not imply strong or coalition-proof equilibrium.

DD-C-0079 proves the full-credit dynamic posterior-mode equilibrium. DD-C-0080 is the exact dynamic planner and response census. DD-C-0081 preserves the visible-action herding null. DD-C-0082 is a separate planner-only threshold-two extension.

DD-C-0083 through DD-C-0086 are exact bounded mechanism results. Their verification uses all 81 labeled action profiles per fixture and an independent 4,050-entry action table. Applicability labels are part of the evidence: an authoritative assignment does not “pass” a voluntary deviation test.

DD-C-0087 is the independently reproduced DiscoveryBench v3 exact matrix. DD-C-0088 is a verified seeded Monte Carlo estimate. The latter has no exactness basis and reports power and intervals to six decimals from rejection counts. These categories are not interchangeable.

C Design checklist for applications

An application should specify at least the following before using the model as a benchmark:

1. the candidate set and posterior or signal channel;
2. the number, skills, and action capacity of agents;
3. the threshold and whether it varies by candidate;
4. execution probability and value above the threshold;
5. costs of stranded attempts, false positives, and redundant members;
6. what information each actor observes and when;
7. whether actions and outcomes are public, delayed, or private;
8. who has assignment authority and whether recommendations bind;
9. the reward, transfer, subsidy, and budget rules;
10. the relevant unilateral and coalitional deviation classes;
11. the equilibrium or learning selection process;
12. the metrics that remain incomparable and therefore unaggregated.

Failing to specify these dimensions invites category errors. A planner value may be reported as an equilibrium. An authoritative assignment may be reported as voluntary obedience. Pair stability may be reported as coalition-proofness. A synthetic power estimate may be reported as a behavioral effect. The repository’s evidence policy treats each of those upgrades as invalid.

D Interpretive cases

Consider first a team with $N = 8$ and threshold $\tau = 2$. Capacity permits four viable teams. If the posterior has four clearly dominant candidates, the planner places a pair on each. A policy that places eight agents on the top candidate spends three team-equivalents without opening any additional target. A policy that sends every agent to a different candidate opens none. Both can be described as coordinated in ordinary language, but only the four-pair portfolio uses capacity efficiently under the frozen objective.

Now consider $N = 8$ and $\tau = 5$. Only one viable team can form. The planner opens the top posterior candidate, and the remaining three agents cannot form a second team. Here common-mode concentration and the planner discovery value coincide. The tie follows from capacity. It does not imply that extra members add value, nor does it generalize to a larger team or smaller threshold.

In a decentralized threshold-two game, two agents may have to move together to create a profitable alternative. Unilateral obedience therefore cannot certify resistance to pair deviations. Conversely, passing every named pair deviation does not test three-person changes or deviations with transfers. The deviation class is part of the result, not a footnote.

In a dynamic setting, an earlier action changes both the posterior and the portfolio state. Following may exploit information but increase crowding. Opposing may diversify action but ignore a strong signal. Joining a singleton can complete a team; starting a singleton can create an option; joining a viable team can be redundant. A sensible policy must condition on all three objects: belief, occupancy, and remaining action capacity.

Finally, compare a central assignment with a token mechanism. Both may support the same two-pair portfolio. The assignment uses authority, so voluntary obedience after commitment is not the relevant test. Tokens preserve action choice but restrict prize eligibility, so obedience and equilibrium multiplicity matter. Reporting only expected discovery would erase the central institutional difference.

E Prediction and measurement boundaries

The synthetic experiment extension maps the theory to registered contrasts. It can ask whether team tokens change viable-team formation, whether visible history changes herding, or whether stopping feedback reduces expected actions. The power calculation requires an assumed effect and response model. It cannot tell whether participants understand the institution, whether identities create fairness concerns, or whether the experimental task represents a real organization.

Outcome measurement should preserve the portfolio decomposition. Discovery alone cannot distinguish a lucky concentrated policy from systematic team formation. Expected distinct actions alone rewards stranded singletons. Useful outcomes include viable-candidate count, posterior mass opened, failed subthreshold attempts, excess overlap, largest crowd, action count, and the distribution of rewards. No one of these is a universal composite score.

The benchmark and experiment package also preserve version boundaries. DiscoveryBench v1 and v2 remain unchanged prefixes of v3. Synthetic Experiment v1 and v2 remain public alongside v3, and shared-seed v2 power rows are exactly preserved. Versioning makes changes auditable without rewriting earlier evidence.

Human deployment remains outside the current program. A future study would need a new issue and registration specifying ethics, consent, privacy, retention, recruitment, exclusion, stopping, identification, and governance. Nothing in the present PDF authorizes that work.

F Planner proof and equilibrium audit detail

The planner proof can be read as an upper bound followed by a construction. For any occupancy n , let $O(n) = \{j : n_j \geq \tau\}$ be the opened set. Since each opened candidate consumes at least τ agents, $|O(n)|\tau \leq N$. Thus $|O(n)| \leq \lfloor N/\tau \rfloor$. The opened set is also a subset of M candidates, so its cardinality is at most K . Posterior ranking then gives

$$D(n; \pi, \tau) = \sum_{j \in O(n)} \pi_j \leq \sum_{j=1}^K \pi_{(j)}. \quad (6)$$

Assigning exactly τ agents to each ranked candidate 1 through K opens that set. The number used is $K\tau \leq N$. Placing the remaining $N - K\tau$ agents anywhere cannot close an opened candidate, so the bound is attained. The argument uses no continuous relaxation and no limiting step.

The DD-016 hand audit tests this theorem on 63 small allocation problems. Three posterior fixtures are crossed with $N = 1$ through 6 and every admissible threshold. Exhaustive enumeration computes the best value independently of the closed form. It also checks normalization and upper and lower bounds. These finite tests do not prove the theorem, but they make common implementation errors hand-detectable: forgetting the minimum with M , assigning a partial team, or treating leftover occupancy as an additional opened candidate.

The payoff identity has a similarly direct finite derivation. For $x_j > 0$, the binomial coefficient relation

$$\frac{1}{k+1} \binom{N-1}{k} = \frac{1}{Nx_j} \binom{N}{k+1} x_j \quad (7)$$

turns the expected equal share at threshold one into $[1 - (1 - x_j)^N]/(Nx_j)$. At threshold two, the $k = 0$ term is excluded, producing the registered subtraction of $(1 - x_j)^{N-1}$. Exact endpoint handling matters: the threshold-one zero limit is one, while the threshold-two zero limit is zero.

The pure-equilibrium verifier does not rely on occupancies alone. The primary method enumerates aggregate states and applies every candidate-to-candidate unilateral comparison. The independent method enumerates labeled action profiles, evaluates every named agent's deviation, and deduplicates the surviving occupancies. Agreement between 3,728 aggregate states and 87,216 labeled profiles protects against multiplicity errors that could be hidden by a single representation.

Coalitional verification changes representation again. The primary evaluator names each coalition and enumerates joint destination actions. The independent method removes a coalition from an occupancy, enumerates possible additions, and solves the exact payoff-compatible matching problem between named members and destination slots. A removed equilibrium, altered best value, and changed pair-stability label are targeted corruption tests. Passing them supports the bounded census, not a claim beyond its registry.

G Institutional audit matrix

The mechanism comparison is easiest to interpret as a matrix of questions. First ask whether the institution controls action. Central assignment and authoritative matching do. Their planner values are allocation results, and a post-commitment deviation test would be a category error. The pairwise matching market binds action within a selected pair but leaves pairs to choose among candidates, so pair-level equilibrium remains relevant.

Second ask whether eligibility changes with identity. Team tokens condition the prize on a fixed balanced token endowment. Exclusive coalition rights name the coalition entitled to the candidate

prize. Both can make a deviation unattractive even when the physical action is available. This can support the intended portfolio, but it requires verifiable identity and a legitimate rule for allocating the scarce rights.

Third ask whether recommendations are private and correlated. The correlated mediator randomizes over balanced teams and sends individual recommendations. Its ex ante anonymity differs from a fixed named assignment. Obedience is a posterior comparison conditional on the recommendation. The bounded result does not show that agents truthfully report private signals to the mediator because the registry starts from a common posterior.

Fourth ask what contribution earns the prize. Universal pooling splits among all agents at the successful viable target. Sole-team rescue conditions payment on having exactly one viable team. Marginal coalition contribution restricts payment to an exactly minimal viable team. Threshold-adjusted equal split uses the ordinary team share at the recommended two-pair portfolio. Similar recommendations can therefore rest on different counterfactual rewards.

Fifth ask which deviations were actually evaluated. The census records unilateral changes, named pair changes, and exact-size-two threshold-team changes where they apply. A block requires strict improvement for every member. It does not allow side payments or count a weak improvement for one member. Those choices preserve a finite exact audit but leave larger coalitions and other stability concepts open.

Finally ask what happens away from the recommendation. The complete 81-profile table for each fixture and mechanism counts all labeled pure equilibria under the registered rule. High multiplicity is not a verification failure. It is an institutional risk that would be lost if the audit reported only whether the planner profile is supported. The correct conclusion is conditional: several rules support a high-discovery team portfolio in the frozen fixtures, while no selection theorem identifies that portfolio as the unique realized outcome.

H Preserved failures and open questions

The evidence program retains failures because they determine what the paper cannot claim. The canonical threshold result includes regimes where common mode raises discovery relative to private clue-following and regimes where it lowers it. Reporting only one side would turn a phase diagram into a slogan. The capacity transition at threshold five is equally important: once only one team can form, the diversified planner cannot improve on a correctly selected common mode in the frozen model.

The equilibrium census retains weak zero-discovery equilibria. They are not removed because they produce an inconvenient infinite price-of-anarchy label. It also retains 21 failures of the tied-mode mixed rule, all at threshold one. Those failures identify exactly why the unsupported-action condition in Equation 5 is necessary.

The dynamic program retains the visibility-herding null. Visible histories strictly lower discovery and dispersion in 18 fixed-budget cells and tie in the other 14; there is no registered cell where they improve either outcome. That result is bounded, but it rules out presenting action visibility as uniformly helpful within the declared grid. The threshold-two extension remains labeled planner-only because no decentralized threshold equilibrium was solved there.

The mechanism program retains the two concentrated failures. Universal pooling and sole-team rescue do not attain the two-team planner value in any of the five fixtures. It also retains multiplicity for mechanisms that support the planner recommendation. A passing obedience check is not converted into uniqueness. Not-applicable truthfulness and authority fields remain not applicable rather than

being counted as favorable results.

The benchmark retains 660 capability exclusions. The experiment package retains 644 large-sample calibration failures. Neither artifact fills missing values with convenient defaults. This makes the public surface less compact but keeps the inference boundary reviewable.

Several questions remain open. Heterogeneous team skills may make a team worth more than the sum of interchangeable members. Candidate-specific thresholds may change posterior ranking into a knapsack problem. Stochastic execution may create value above the minimum team size. Endogenous source acquisition may couple who learns with who acts. Private posteriors require an elicitation and truthfulness layer absent from DD-018. Larger coalitions and transfers may overturn pairwise stability. Learning dynamics may select equilibria that the static census merely lists.

These extensions should not be answered by stretching the present labels. Each requires a new registered model, resource estimate, verifier, and evidence category. A new numerical claim would require a new immutable run; a new theorem would require a proof and proportional tests. Until then, they remain questions rather than implied conclusions.

References

- Robert J. Aumann. Acceptable points in general cooperative n-person games. In *Contributions to the Theory of Games IV*, volume 40 of *Annals of Mathematics Studies*, pages 287–324. Princeton University Press, 1959.
- B. Douglas Bernheim, Bezalel Peleg, and Michael D. Whinston. Coalition-proof Nash equilibria i. concepts. *Journal of Economic Theory*, 42(1):1–12, 1987. doi: 10.1016/0022-0531(87)90099-8.
- Anna Bogomolnaia and Matthew O. Jackson. The stability of hedonic coalition structures. *Games and Economic Behavior*, 38(2):201–230, 2002. doi: 10.1006/game.2001.0877.
- Andreas Diekmann. Volunteer’s dilemma. *Journal of Conflict Resolution*, 29(4):605–610, 1985. doi: 10.1177/0022002785029004003.
- Christian Hellwig and Laura Veldkamp. Knowing what others know: Coordination motives in information acquisition. *The Review of Economic Studies*, 76(1):223–251, 2009. doi: 10.1111/j.1467-937X.2008.00515.x.
- Philip Kitcher. The division of cognitive labor. *The Journal of Philosophy*, 87(1):5–22, 1990. doi: 10.2307/2026796.
- James G. March. Exploration and exploitation in organizational learning. *Organization Science*, 2(1):71–87, 1991. doi: 10.1287/orsc.2.1.71.
- Jacob Marschak and Roy Radner. *Economic Theory of Teams*. Number 22 in Cowles Foundation Monograph. Yale University Press, New Haven, 1972. URL <https://cowles.yale.edu/research/cfm-22-economic-theory-teams>.
- David P. Myatt and Chris Wallace. Endogenous information acquisition in coordination games. *The Review of Economic Studies*, 79(1):340–374, 2012. doi: 10.1093/restud/rdr018.
- Yohei Nakajima. The shared discovery paradox: How a one-answer rule turns better information into worse search. Working paper, 2026. URL <https://github.com/yoheinakajima/shared-discovery-paradox>. Canonical source pinned at commit 5025cc8e8f2f8ca015dff2066f08f81ad5715a51.

- Thomas R. Palfrey and Howard Rosenthal. Participation and the provision of discrete public goods: A strategic analysis. *Journal of Public Economics*, 24(2):171–193, 1984. doi: 10.1016/0047-2727(84)90023-9.
- Robert W. Rosenthal. A class of games possessing pure-strategy Nash equilibria. *International Journal of Game Theory*, 2:65–67, 1973. doi: 10.1007/BF01737559.
- Fiona Y. Wang, Lee Marom, Subhadeep Pal, Rachel K. Luu, Wei Lu, Jaime A. Berkovich, and Markus J. Buehler. Autonomous agents coordinating distributed discovery through emergent artifact exchange. arXiv:2603.14312, 2026.
- Kevin J. S. Zollman. The communication structure of epistemic communities. *Philosophy of Science*, 74(5):574–587, 2007. doi: 10.1086/525605.

Artifact declaration

This document is a project-authored working paper distributed under the repository’s MIT license. It has no DOI, has not been submitted, and has not been peer reviewed. It makes no verified novelty claim. The bibliography positions the model against established literatures; it is not a novelty certificate.

The authoritative numerical sources are these immutable runs:

- 20260722T021526Z_DD-016_00271ff8_123b2809e3;
- 20260722T024032Z_DD-017_033452f6_3d2c74fdfb;
- 20260722T043713Z_DD-015_92d53ac1_0e7cf1ec0a;
- 20260722T044453Z_DD-015_34bc4379_33e1da478b;
- 20260722T051847Z_DD-018_a193f602_3b3ddac173;
- 20260722T054447Z_DD-010_d265e480_6930915b02; and
- 20260722T061958Z_DD-011_5743ccba_19b6517655.

They must not be rerun for freshness.

Reproduction requires the pinned repository dependencies, Tectonic, and Poppler. `makethreshold-discovery` validates input checksums, regenerates eight assets, resolves claim and citation keys, compiles twice under a fixed source epoch, rejects unresolved references and overfull boxes, and requires byte- identical PDFs. Any source or PDF checksum change invalidates the visual review and requires a complete rebuild and all-page inspection.

No participants were recruited. No human data were collected. No experiment was conducted.